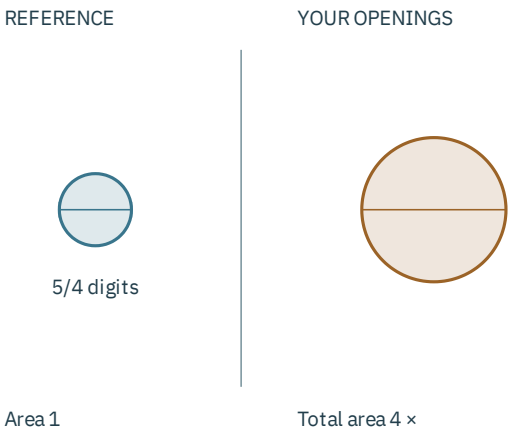
Double the bore.
Four times the opening.

A Roman diameter name, a circular area, and a separate question about delivery.

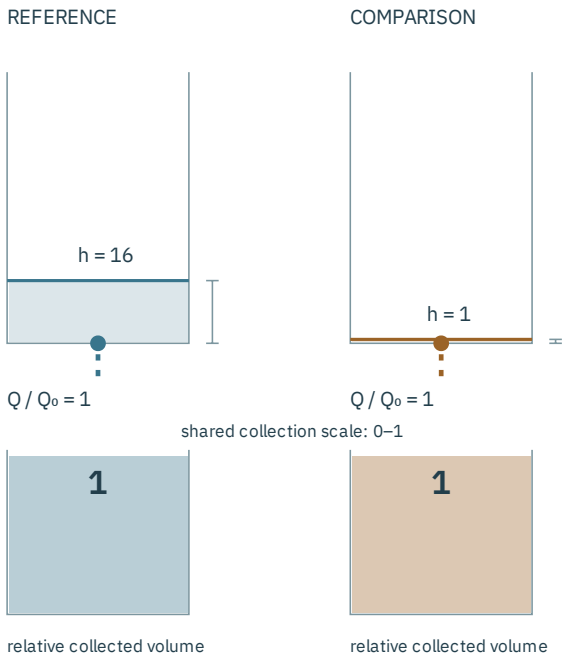
Reopen these settings and the trial book. This script-free file embeds six full source pages, nine readings, diagrams, your trial book, three fonts and complete licenses. It makes no external subresource requests.

Your ideal-opening comparison

One reference opening has diameter 5/4 source digits and head 16 chosen units. Your 1 opening each have diameter 5/2 source digits and head 1. Area ratio 4; head ratio 1/16; relative delivery 1. Over 1 chosen time units the relative volumes are 1 and 1.



Common length scale. Separate circles compare areas; they do not show circle packing.



Constant heads, common coefficient, no pipe losses. The two collection bars use a shared linear scale. The visible reservoir widths do not define those heads or volumes.

For circular openings A/A_0 = count × $(D/D_0)^2$. The modern ideal-orifice relation $Q = C\,A\,\sqrt{(2gh)}$ gives $Q/Q_0 = (A/A_0)\,\sqrt{(h/h_0)}$ when C and g are common. Squared flow ratio: 1 exactly. Displayed square roots are rounded.

This is a modern ideal-orifice ratio experiment prompted by Frontinus's distinction between gauge, opening and delivery. It is not a reconstruction of a Roman calix or a prediction for a real supply pipe. Both reservoirs maintain constant head, the outlet coefficients are equal, approach speed and pipe losses are omitted, and every time and head unit is chosen for the comparison. No source digit is converted to metres and no quinaria is converted to litres per second. The printed ancient capacities, the geometric opening ratios and the model's relative flow remain different quantities.

Modern physical derivation: OpenStax, Bernoulli's equation.

The recorded trial book

Recorded predictions · one constant-head trial per row

Trial	D / digits	Openings	Head	Area / A_0	Q / Q_0	Time	Volumes A / B
1	5/2	1	16	4	4	1	1 / 4
2	5/4	4	16	4	4	1	1 / 4
3	5/2	1	1	4	1	1	1 / 1
4	5/4	1	64	1	2	1	1 / 2

These are computed predictions, not observations of ancient equipment. Time and relative-volume units are chosen and have no conversion to seconds or litres.

Nine selected Latin readings

Latin follows Bücheler's 1858 printed edition, with joined line endings, editorial apparatus omitted from the excerpt, and u/v spelling retained. I.43 explicitly omits circumference figures with an ellipsis. English translations and language notes are new teaching aids. These are selections, not a complete transcription. Full printed pages preserve the apparatus and context.

Frontinus I.25 · printed p. 014

1. A probable name

maxime probabile est, quinariam dictam a diametro quinque quadrantum

It is most probable that the quinaria takes its name from a diameter of five quarters.

maxime probabile · most probable. An explicit judgement about the name's origin. The author does not present a directly observed invention.

quinque quadrantum · of five quarters. Quadrantum is a genitive plural. The diameter is five quarter-digits: 5/4 of one digit.

Five quarters of a digit, not five quinariae. Frontinus presents this as the most probable of competing explanations. He has just questioned the derivation from a five-digit-wide lead sheet. The diagram follows his stated diameter rule, rather than claiming a measured surviving pipe.

Frontinus I.25 · printed p. 014

2. Six names a diameter

diametro per singulos adiectione singulorum quadrantum crescente: ut in senaria, quae sexs quadrantes in diametro habet

The diameter increases from one size to the next by adding a quarter each time: for example, the senaria has six quarters in its diameter.

diametro … crescente · with the diameter increasing. An ablative construction: the increasing diameter describes how this named series is generated.

senaria · the six-pipe. The printed sexs is a spelling of sex, six. Six counts quarter-digits across this opening. It does not mean that the opening equals six quinaria openings.

Increasing the diameter from five quarters to six quarters multiplies it by 6/5. In a circular opening the area multiplies by $(6/5)^2 = 36/25$. That geometric result is distinct from an absolute flow rate and from the source's edited capacity table.

Frontinus I.26 · printed p. 014

3. Three descriptions of one opening

omnis autem modulus colligitur aut diametro aut perimetro aut areae mensura, ex quibus et capacitas adparet

Each gauge is determined by its diameter, its circumference, or its area measure; from these its capacity is also apparent.

modulus · gauge, standard measure. The word refers here to a size or standard used in distribution, rather than a modern software module.

areae mensura · measure of the area. Areae is a genitive. Area measures the two-dimensional opening; a collected volume also needs a length dimension, supplied through water moving over time.

For a circle, any one of those geometric measures determines the other two. This sentence does not supply a universal modern volume-per-time conversion. The later pressure passage is retained beside it.

Frontinus I.27 · printed p. 015

4. Several quinariae in one opening

est unum, cum ipsa multiplicatur, id est eodem lumine plures quinariae includuntur, in quibus secundum adiectionem quinariarum amplitudo luminis crescit

One method multiplies the quinaria itself: several quinariae are included in the same opening, whose size increases as quinariae are added.

lumine · in the clear opening. Lumen can mean light. In this pipe context it identifies the open passage through which water can pass.

amplitudo luminis · size of the opening. This grows by adding opening areas. The diameter grows as the square root of that count, not in direct proportion to it.

The argument compares total opening, not how many separate circles can be packed inside another circle. Four quinaria-sized areas equal the area of a circle with twice the diameter. Four actual small circles cannot simply be fitted inside that larger circle without gaps or overlap.

Frontinus I.28 · printed p. 015

5. A different way to increase

ut puta quinaria, cum adiectus est ei ad diametrum quadrans, senariam facit. nec iam in solidum capacitatem ampliat

For example, adding a quarter to the quinaria's diameter makes a senaria. This does not increase its capacity by a whole unit.

ad diametrum · to the diameter. Ad with the accusative identifies the dimension receiving the quarter. It is not an addition of a quarter to the area.

in solidum · by a whole unit. The following source gives a fractional capacity. This workshop derives a circular area ratio without silently replacing the edition's fractional table.

The name changes by one; the geometric opening does not increase by one whole quinaria-sized area. The source also uses a different area-based naming scheme above the twenty-pipe. This workshop limits its adjustable diameter to five through ten quarter-digits, so that it does not mix those schemes.

Frontinus I.35 · printed p. 017

6. A higher incoming supply

Meminerimus omnem aquam, quotiens ex altiore loco uenit et intra breue spatium in castellum cadit, non tantum respondere modulo suo set etiam exuberare

Let us remember that water arriving from a higher place and falling into a distribution tank over a short distance can more than answer to its gauge.

Meminerimus · let us remember. A first-person plural form of memini used here as an instruction to keep this qualification in view.

castellum · distribution tank. In this water-supply context the word identifies a reservoir or distribution structure, not a military castle.

The author explicitly brings height and the supply route into the account. Our model holds two reservoir surfaces constant and changes one opening's depth. That supplies a controlled modern comparison; it does not compute Frontinus's complete incoming pipe or tank arrangement.

Frontinus I.35 · printed p. 017

7. Less pressure, a longer route

quotiens uero ex humiliore, id est minore pressura, longius ducitur, segnitia ductus modum quoque deperdere

When it is conducted farther from a lower place—that is, with less pressure—the sluggish conveyance also loses some of its amount.

minore pressura · with less pressure. Both words are ablative singular. Minor is comparative: less, relative to a more strongly driven supply.

longius ducitur · it is conducted farther. The passive ducitur describes conveyance. A real longer pipe introduces effects absent from the ideal-orifice model.

Minore pressura is part of the ancient wording. The ideal-orifice calculation here omits pipe friction and length; it models only the separate effect of head under common assumptions. Frontinus's warning must not be turned into a claim that geometry alone fixes delivery.

Frontinus I.36, continued · printed p. 018

8. The calix is a bronze fitting

est autem calix modulus aeneus, qui riuo uel castello induitur; huic fistulae adplicantur. longitudo eius habere debet digitos non minus duodecim

The calix is a bronze gauge inserted into the conduit or distribution tank; pipes are attached to it. Its length must be at least twelve digits.

modulus aeneus · a bronze gauge. Aeneus names the material. The source goes on to connect bronze's resistance to bending with keeping a controlled opening.

non minus duodecim · not less than twelve. This is a minimum length in source digits. It is not the diameter of a twelve-pipe and supplies no conversion of a quinaria into litres per second.

The source's calix has a stated length and attached pipes. Our mathematical openings have no modeled tube length. A thin ideal orifice is therefore a counterexample showing why more conditions are needed, rather than an archaeological or hydraulic reconstruction of this fitting.

Frontinus I.43 · printed p. 019

9. The ten-pipe holds four

Fistula denaria: diametri digitos duos et semis … capit quinarias IIII.

The ten-pipe: a diameter of two and a half digits … it holds four quinariae.

duos et semis · two and a half. Two and a half digits are ten quarter-digits. Doubling a circular diameter quadruples the opening's area.

capit quinarias IIII · it holds four quinariae. IIII writes four. Keep the ancient reported capacity distinct from the experiment's geometric area ratio and chosen relative-flow unit.

The omitted words give its circumference. The source's ten-pipe doubles the diameter of the five-quarter quinaria and reports four quinariae. The modern circle calculation agrees for this example. This agreement does not settle all damaged or edited fractions elsewhere in the table; the complete page and apparatus remain available.

Read the qualification beside the number

The ten-pipe in I.43 has a diameter twice that of the quinaria and a reported capacity of four quinariae. Equal circular area does not establish equal real delivery at different pressures, lengths or intake arrangements. Frontinus himself discusses height, pressure and route in I.35 and a bronze calix at least twelve digits long in I.36. Our ideal opening has no modeled tube length.

For the separate four-opening example, four equal small areas sum to one area four times as large. That is an area equality, not a claim that four circles fit inside a circle twice their diameter. The name-to-diameter series is limited here to five through ten quarter-digits; the source uses another naming scheme for larger pipes. The capacity table contains editorially disputed fractions. This experiment does not silently repair them or make every edited figure conform to modern geometry.

Compare the Latin Library transcription and Bennett's 1925 English translation. Our retained printed witness is Bücheler, Leipzig: B. G. Teubner, 1858.

Continue with Hero's Greek argument about collecting water and observing time. This is a comparison of two texts, not a claim that one formula appears in either ancient source.

Printed p. 014 · Names, diameters and areas

Complete uncropped page · PDF page 35, archive leaf 34. Source volume at this leaf.

anguli deteruntur. postea modulus nec ab uncia nec ab alterutro digitorum originem accipiens inductus, ut quidam putant, ab Agrippa, ut alii, a plumbariis per Vitruuium architectum in usum urbis exclusis prioribus uenit, appellatus quinariae nomine. qui autem Agrippam auctorem faciunt, dicunt, quod quinque antiqui moduli exiles et uelut puncta, quibus olim aqua cum exigua esset diuidebatur, in unam fistulam coacti sint; qui Vitruuium et plumbarios, ab eo quod plumbea lammina plana quinque digitorum latitudinem habens circumacta in rotundum hunc fistulae modulum efficiat. set hoc incertum est, quoniam cum circumagitur, sicut interiore parte adtrahitur, ita per illam, quae foras spectat, extenditur. maxime probabile est, quinariam dictam a diametro quinque quadrantum, quae ratio in sequentibus quoque modulis usque ad uicenariam durat, diametro per singulos adiectione singulorum quadrantum crescente: ut in senaria, quae sexs quadrantes in diametro habet, et septenaria, quae septem, et deinceps simili incremento usque ad uicenariam. omnis autem modulus colligitur aut diametro aut perimetro aut areae mensura, ex quibus et capacitas adparet. differentiam unciae, digiti quadrati et digiti rotundi, et ipsius quinariae ut facilius dinoscamus, utendum est substantia quinariae, qui modulus et certissimus et maxime receptus est. unciae ergo modulus habet diametri digitum unum et trientem digiti; capit plus, quam *quinaria*, quinariae octaua, hoc est sescuncia quinariae et scripulis tribus et besse scripuli. digitus quadratus in rotundum redactus habet diametri digitum unum et digiti sescunciam sextulam; capit quinariae dextantem. digitus rotundus habet diametri digitum unum; capit quinariae septuncem semunciam sextulam. ceterum moduli,

1 detēruntur *C* | 3 agryppa *item v.* 5 agryppam *C* | plumbaris *C* | uitrinnum *C* | 4 quinariae *scripsi* quinario *C* | 7 sint *Heinrichius* sunt *C* | 11 adtraitur *C* | 14 uicinariam *C* | adiectionem *C* | 15 ud *C* | sex. s. *C* | 18 aud diametro *C* | p̃erymetro *C* | auderee mensure *C*, *em. Polenus* | 19 abparet *C* | 20 ipsi *C* | ud *C* | 21 substantie *C*, *em. Polenus* | 22—28 *foede corrupta repetuntur cap.* 38 | 23 quinaria *addidi* capit quinariam et plus quam quinariae octauam, hoc est sescunciam quinariae et scripula tria et bessem scripuli *Polenus* | 24 scse uncia *C*, *em. Iocundus* | 25 bese *C* | 26 sexcunciam *C* | sextulam *Iocundus* sextum *C* scripulum *accuratiore computandi ratione usus Polenus* | 27 dextantem]

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Printed p. 015 · Two ways to enlarge a gauge

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qui a quinaria oriuntur, duobus generibus incrementum accipiunt. est unum, cum ipsa multiplicatur, id est eodem lumine plures quinariae includuntur, in quibus secundum adiectionem quinariarum amplitudo luminis crescit. est autem fere tum in usu, cum plures quinariae impetratae, ne riuus saepius conuulneretur, una fistula excipiuntur in castellum, ex quo singuli suum modum recipiunt. alterum genus est, quotiens non ad quinaria- 28
rum necessitatem fistula incrementum capit, set ad diametri sui mensuram, secundum quod et nomen accipit et capacitatem ampliat: ut puta quinaria, cum adiectus est ei ad diametrum quadrans, senariam facit. nec iam in solidum capacitatem ampliat; capit enim quinariam unam et quincuncem sicilicum. et deinceps eadem ratione quadrantibus diametro adiectis, ut supra dictum est, crescunt septenaria, octonaria usque ad uicenariam. subsequitur illa ratio, quae constat ex numero digitorum quadra- 29
torum, qui area, id est lumine, cuiusque moduli continentur, a quibus et nomen fistulae accipiunt. nam quae habet areae, id est luminis in rotundum coacti, digitos quadratos uiginti quinque, uicenum quinum adpellatur: similiter tricenaria et deinceps pari incremento quinorum digitorum quadratorum usque ad centenum uicenum. in uicenaria fistula, quae in confinio utriusque rationis 30
posita est, utraque ratio paene congruit. nam habet secundum eam computationem, quae in antecedentibus modulis seruanda est, † in diametro quadrantes uiginti, cum diametri eiusdem digiti

dodrantem semunciam sicilicum *Polenus* | 1 quinarie *C* | 2 Est *Heinrichius* et *C* | unum *Polenus* una *C* | multiplicantur *C* | iumine *C* | 4 tum *scripsi* īt *C* tunc *cod. Vrb.* | 5 ne riuus *Iordansius de publ. urb. R. et Constantinop. aquaeduct. p.* 55 neminis *C* | septus *C, em. Iocundus* | conuul'neretur *C* | 8 necessitate *C* | 9 capacitatem *Iocundus* capacitatis *C* | 11 senarium *C, em. Schultzius* | 12 sycylycum *C* | 13 ud *C* | 16 id est lumine *suspecta* | continentur *Polenus* continetur *C* | 19 abpellatur *C* | pari incremento quinorum *scripsi* per incrementorum *C* per incrementum *cod. Vrb.* | 21 quae *an* quod *in C sit non liquet* | 22 utraque] ut ˜que˜ *C* | 22—16, 2 *in hanc sententiam restituenda:* nam habet secundum eam computationem, quae in antecedentibus modulis seruanda est, aream digitorum quadratorum exiguo minus uiginti, cum diametri eiusdem digiti quinque sint; et secundum eorum modulorum rationem, qui secuntur, aream habet digitorum quadratorum uiginti | 22 habes *C* | 23 in tecedentibus *C, em. Polenus*

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Printed p. 017 · Pressure and intake position

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quam tredecim, et ex centenaria, quam ampliauerunt eque *centenum uicenum* certum est illos non erogare nisi ad artiorem numerum, quia Caesar secundum suos commentarios, cum ex quaque centenaria expleuit quinarias octoginta unam semissem, item ex centenum uicenum quinarias nonaginta octo, tamquam exhausto modulo desinit distribuere. In summa moduli sunt XX quinque. 34 omnes consentiunt et rationi et commentariis exceptis his quattuor, quos aquarii nouauerunt. omnia autem quae mensura continentur, certa et immobilia congruere sibi debent; ita enim uniuersitati ratio constabit. et quemadmodum uerbi gratia sextarii ratio ad cyatos, modii uero et ad sextarios et ad cyatos respondet: ita et quinariarum multiplicatio in amplioribus modulis seruare consequentiae suae regulam debet. alioqui cum in erogatorio modulo minus inuenitur, in acceptorio plus, adparet non errorem esse set fraudem.

Meminerimus omnem aquam, quotiens ex *altiore* loco uenit 35 et intra breue spatium in castellum cadit, non tantum respondere modulo suo set etiam exuberare; quotiens uero ex humiliore, id est minore pressura, longius ducitur, segnitia ductus modum quoque deperdere; et ideo secundum hanc rationem aut onerandam esse erogatione aut releuandam.

Set et *calicis* positio habet momentum. in rectum et ad 36 libram conlocatus modum seruat, ad cursum aquae obpositus et deuexus amplius rapit, ad latus praetereuntis aquae inuersus et supinus, id est ad haustum pronior, segniter et exiguum sumit.

1 eque centenum uicenum *scripsi* eque *quod* aeque *interpretabantur C* ex centenaria et centenum uicenum quas ampliauerunt *Schultzius* | 4 semissem *Polenus* se ..... *C* | 5 LXXXXVII et dodrantem *Polenus* | tamquam *uel* tumquam *C* | 6 XX] moduli *quod primo scripserat eadem manus deleuit et* XX *superposuit in C* | 8 continetur *C* | 10 Ed *C* | 11 respondet *scripsi* respondent *C* | 14 acceptore *O* | abparet *C* | 16 Memineramus *C* | ex altiore loco *Pithoeus* ex ..... o *C* | 17 s. at cum *C, em. Iocundus* | 18 Quotien' *C* | 19 ducitur *scripsi* ducatur *C* | 20 deperderet *C, em. Schultzius* | oneranda *C, em. ed. Basil.* | 21 erogationem *ed. Basil.* | releuanda *C, em. ed. Basil.* | 22 calicis *add. Iocundus* | 23 libram *C* | seruat *Iocundus* seruauit *C* | 24 īuersus *C cui cum* cuersus *simillimum sit* conuersus *cod. Vrb.* | 25 *malim* id est — pronior *ad* obpositus et deuexus *interpretandi caussa adscriptum iudicare* | segniter et *scripsi* segniterer *C* segniter *cod. Vrb.*

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Printed p. 018 · The bronze calix and the table

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est autem calix modulus aeneus, qui riuo uel castello induitur; huic fistulae adplicantur. longitudo eius habere debet digitos non minus duodecim, lumen [id est] capacitatem quanta imperata fuerit. excogitatus uidetur, quoniam rigor aeris difficilior ad flexum non temere potest laxari uel coartari.

37 Formulas modulorum qui sunt omnis uiginti et quinque subieci, quamuis in usu quindecim tantum frequentes sint, derectas ad rationem de qua locuti sumus, emendatis quattuor, quos aquarii nouauerant. secundum quod et fistulae omnes, quae opus facient, derigi debent aut si haec fistulae manebunt, ad quinarias quot capient conputari. qui non sint in usu moduli, in ipsis est adnotatum.

38 [*Vncia* habet diametri *digitum unum et* trientem digiti; *capit plus*, quam quinaria, quinariae sescuncia scripulis tribus et besse scripuli. [digitus quadratus in latitudine et longitudine aequalis est.] digitus quadratus in rotundum redactus habet diametri digitum unum et digiti sescunciam *sextulam*, capit quinariae *dextantem*. digitus rotundus habet diametri digitum unum, capit quinariae septuncem et semiunciam sextam.]

39 Fistula quinaria: diametri digitum unum ⚌—, *perimetri* digitos tres S⚌⚌—ↄ III, capit quinariam unam.

40 Fistula *senaria:* diametri digitum unum semis, perimetri digitos IIII S⚌℔ↄ II, capit quinariam I⚌⚌—ↄ VII.

1 uel] ut *C* | inducitur *Fea storia delle acque p.* 227 *et p.* 347 | 2 abplicantur *C* | 3 lumen] lm̃n *C* | id est *seclusi* eius *Schultzius* | capacitate *C* | 5 timeri *C* | coortari *C* | 6 Formlas *C* | om̃s *C* | 7 subieci *Schultzius* subiecti *C* | directas *Schultzius* derectam *C* | 9 quod] quam *ed. Florentina deleto* et | 10 aud *C* | hec *an* hee *in C sit non omnino certum* | 11 quod *C* | 13 *e cap.* 26 *huc translata iudicauit Schultzius* | Ed diametri trientem digitum dici. quam qui quinarie *C* | 15 besscripuli *C* | 17 digitu *C* | sescuncia capit quinarie. Digitus *C* | 19 septuncen *C* | 20 '*desideramus uerbum* habet.' *Dederichius* | quadrantem, perimetri *add. Polenus* | 21 tres ↄℰZZ rm *C*, *em. Polenus* | quinaria una *C*, *em. Iocundus* | 22 senaria *add. Iocundus* | 23 IIII S⚌℔ↄ II *scripsi* uu. SZ⅃ↄш *C* quattuor, bessem, semunciam, scripulum *Polenus* | quinariam I⚌⚌—ↄ VII *scripsi* quinarias nouem. ZZZↄ *C* quinariam unam, quincuncem, sicilicum *Polenus. cf. cap.* 28

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Printed p. 019 · The ten-pipe and edited fractions

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Fistula septenaria: diametri digitum I S ⁝·, perimetri digitos V S, capit quinariam I S ⚌⚌— ℒ; in usu non est. 41

Fistula octonaria: diametri digitos duos, perimetri digitos sex ⚌— ϶ X, capit quinarias II S ℒ ϶ quinque. 42

Fistula denaria: diametri digitos duos et semis, perimetri digitos septem S ⁝⁝ ϶ VII, capit quinarias IIII. 43

Fistula duodenaria: diametri digitos III, *perimetri digitos* VIIII ⚌⚌— ϶ III, capit quinarias quinque S ⚌— ϶ III; in usu non est. [alia] aput aquarios habebat diametri digitos III ℒ ϶ VI, capacitatis quinarias sex. 44

Fistula quinum denum: diametri digitos III S ⚌—, perimetri digitos XI S —⚌ ϶ X, capit quinarias nouem. 45

Fistula uicenaria: diametri digitos quinque ℒ ϶, perimetri digitos XV S ⁝⁝ ϶ VI, capit quinarias sedecim ⚌— ℒ. aput aquarios habebat diametri digitos IIII S, capacitatis quinarias XIII. 46

Fistula uicenum quinum: diametri digitos quinque S · ℒ ϶ V, perimetri digitos decem et septem S ⚌ ℒ ϶ VII, capit quinarias XX ⚌⚌ ϶ VIIII; in usu non est. 47

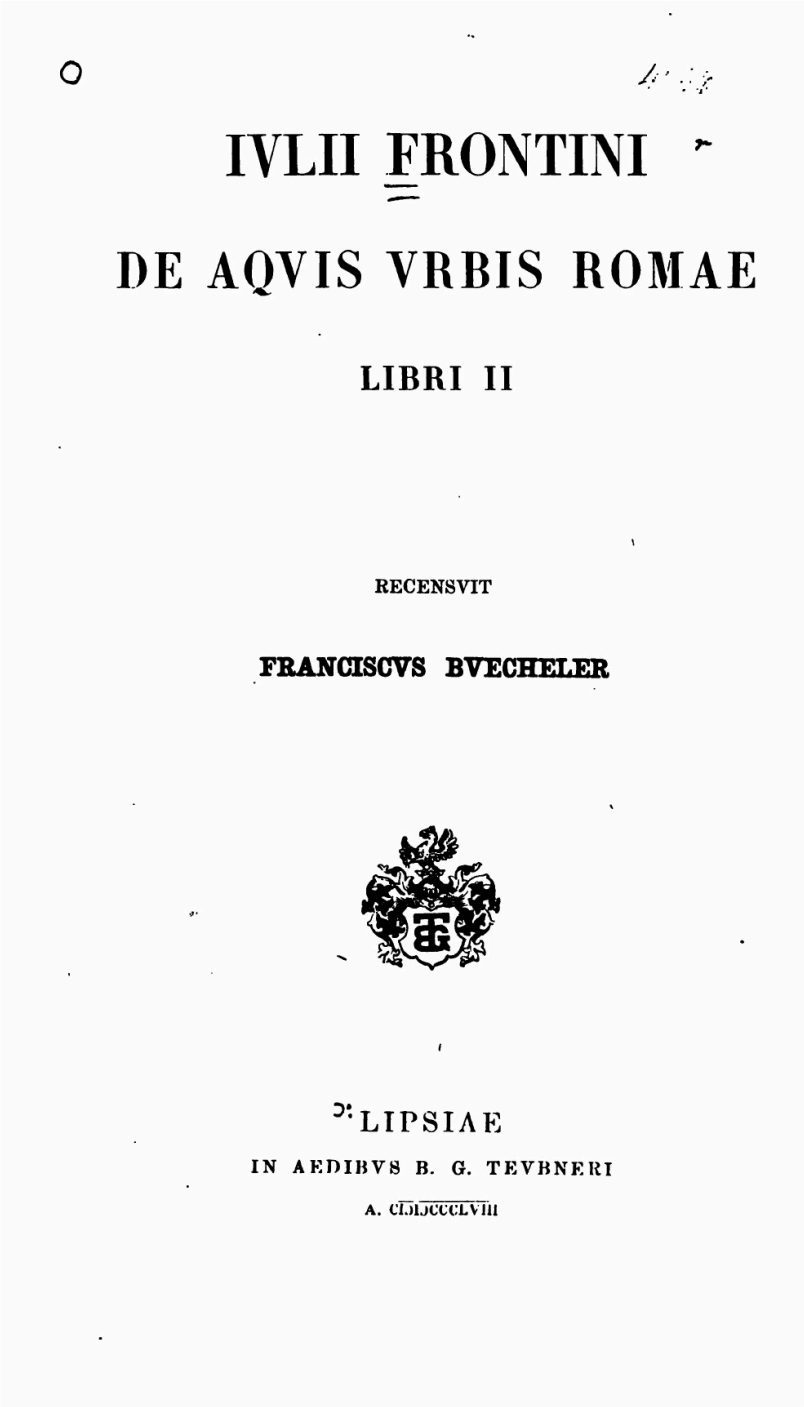
Fistula tricenaria: diametri *digitos* sex ⚌ ϶ III, perimetri di- 48

1 ι S ⁝· *C, item Polenus* | 2 V, semissem *Polenus* sex *C* | quinariam unam, deuncem, semunciam *Polenus* quinarias. l. *C* | 3 sex ⚌— ϶ X *scripsi* sex *C* sex, quadrantem, duellam *Polenus* | 4 quinarias ⊔ ζ ζ ϶ quinque *C* quinarias duas, semissem, semunciam, sicilicum *Polenus* | 6 septem S ⁝⁝ ϶ VII *scripsi* septem ·⁝ ϶ ⊔IIII *C* septem, dextantem, sicilicum *Polenus* | IIII] un *or* *C* | 7 tres, perimetri digitos *add. Polenus* | 8 VIIII ⚌⚌— ϶ III *scripsi* sex. ⚌⚌— ϶ *C* VIIII, quincuncem, scripula II *Polenus* | quinque S ⚌— ϶ III *scripsi* quinque. Σ Z — ϶ *C* V, dodrantem *Polenus* | 9 aquarias abebat *C* | tres semunciam, sicilicum *Polenus* III. S ϶ Y *C* | 11 tres, dodrantem *Polenus* IIII. *or* Z — *C* | 12 XI S —⚌ ϶ X *scripsi* duodecim —⚌. l. Alia *C* XI dodrantem, duellam *Polenus* | 13 quinque ℒ ϶ *scripsi* quinque. Σ 1 *C* quinque *Polenus* | 14 XV S ⁝⁝ ϶ VI *scripsi* XII· S ⁝ ϶ ω S *C* XV, bessem, semunciam *Polenus* | sedecim ⚌— ℒ *scripsi* sedecim. Z — *C* XVI *Polenus* | 15 IIII S *cum Poleno scripsi* octo. Z *C* | quinas *C* | XIII *scripsi* ·mi *C* XII, deuncem, semunciam *Polenus* | 16 quinque ϟ ·ϟ ϶ ⊔ (*id est* ϶ V) *C, item* (V, septuncem, semunciam, sextulam, scripulum) *Polenus* | 17 septem S ⚌ ℒ ϶ VII *scripsi* septem. Σ Ɛ ϶ ⊔ι. *C* XVII, bessem, semunciam, sicilicum *Polenus* | 18 XX ⚌⚌ ϶ VIII *scripsi* XX ⚌⚌ ζ ϶ IIII *C* XX, trientem, semunciam *Polenus* | 19 daametri sex. Z ϶ III *C* VI, sextantem, sextulam *Polenus*

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Title page · Bücheler’s 1858 title page

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